

AI Model with Road Network Planning and Autonomous Driving

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1. Course Content
2. Preparation Work
 - 2.1 Start `kilted_agent` Agent
 - 2.2 Prerequisite Knowledge
3. Run Case Program
 - 3.1 Start Road Network Navigation
 - 3.2 Start Autonomous Driving
 - 3.3 Start multi_brains Agent
 - 3.4 Demonstration Case

1. Course Content

- This course is a comprehensive advanced case study that builds on [13.Road Network Planning with Traffic Sign Recognition] to combine AI large models for more intelligent navigation. It can also combine multiple scenario cases to let AI take over autonomous driving navigation and complete complex cases in the track map.

[!CAUTION]

PI5 mainboard does not support this case!!

2. Preparation Work

2.1 Start `kilted_agent` Agent

- Use the following command to start the microROS agent and connect to the chassis

[!WARNING]

If you use the `start_agent` command to start the microROS agent, you need to close the previous agent first. Note that this is different from the previous microROS agent startup command

```
start_kilted_agent
```

```
jetson@yahboom: ~
jetson@yahboom: ~ 80x24
ubsubscriber created | client_key: 0x14F22590, subscriber_id: 0x000(4), partici
pant_id: 0x000(1)
[1763535321.445792] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1763535321.449556] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x14F22590, topic_id: 0x006(2), participant_
id: 0x000(1)
[1763535321.452876] info | ProxyClient.cpp | create_subscriber | s
ubscriber created | client_key: 0x14F22590, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1763535321.457712] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1763535321.461959] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x14F22590, topic_id: 0x007(2), participant_
id: 0x000(1)
[1763535321.466141] info | ProxyClient.cpp | create_subscriber | s
ubscriber created | client_key: 0x14F22590, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1763535321.473112] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x14F22590, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

- Start ros2kilted container (if already started, no need to repeat)

```
bringup_ros2kilted
```

```
jetson@yahboom: ~
jetson@yahboom: ~ 102x39
jetson@yahboom:~$ bringup_ros2kilted
access control disabled, clients can connect from any host
jetson@yahboom:~$
: Container ros2_kilted-ros2_kilted-1 Starting
1.4s
```

- Start Dify, all functions related to AI large models need to start Dify (if already started, no need to repeat)

```
bringup_dify
```

[!WARNING]

RDk mainboard requires additional execution

Open ~/.bashrc, uncomment this line, and change it to the communication method as shown in the figure (after learning the road network planning function, please restore this comment to avoid affecting previous cases)

```
vim ~/.bashrc
```

```

133 export ROS_DOMAIN ID=16
134 export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
135 #unset RMW_IMPLEMENTATION
136 export ROBOT_TYPE="ROSMaster-M3"
137 export MULTI_ROBOT="" #robot1 or robot2
138 #=====雷达和相机型号=====
139 #=====Radar and camera models=====
140 #=====
141 #单雷达: single          双雷达: dual
142 #Single radar: single    Dual radar: dual
143 #export LASER_TYPE=single
144 export LASER_TYPE=dual
145 #=====
146 #相机类型    dabai_dcw2          as_hp60c
147 #Camera type dabai_dcw2          as_hp60c
148 #export CAMERA_TYPE=dabai_dcw2
149 export CAMERA_TYPE=as_hp60c
150 #=====
151 #export MACHINE=OrinNano
152 export MACHINE=RDkx5

```

```

:wq #Save and exit
source ~/.bashrc #Update environment

```

After environment update, the terminal display changes from green to red

```

IP_Address_1: 192.168.11.80
IP_Address_2: 172.17.0.1
MACHINE: RDkx5 | ROS_DOMAIN_ID: 16 | ROS_DISTRO: humble | RMW
ROBOT_TYPE: ROSMaster-M3 | LASER_TYPE: dual | CAMERA_TYPE: as_hp60c
-----
sunrise@ubuntu:~$

-----
sunrise@ubuntu: ~ 80x11
[System Information]
-----
IP_Address_1: 192.168.11.80
IP_Address_2: 172.17.0.1
MACHINE: RDkx5 | ROS_DOMAIN_ID: 16 | ROS_DISTRO: humble | RMW
ROBOT_TYPE: ROSMaster-M3 | LASER_TYPE: dual | CAMERA_TYPE: as_hp60c
-----
sunrise@ubuntu:~$

```

2.2 Prerequisite Knowledge

For a better experience, please first learn and master the following content in the product tutorial:

- AI Large Model: [3.AI Model Basics], [4. AI Model Development], [5.AI Model-Voice Version]
- Road Network Planning Navigation: [7.Road Network Planning and Navigation]
- Autonomous Driving: [01.Environment Setup & Precautions], [07.Vehicle Centerline Calibration], [08.Lane Detection], [09.Lane-Keeping Autonomous Driving], [10.Traffic Sign Recognition], [14.Road Network Planning with Traffic Sign Recognition]

3. Run Case Program

[!IMPORTANT]

- Need to build the track map's grid map: map, pose map: pose_map, and road network description file: roadnet_file in advance according to [08.Autonomous Driving expand(Need purchase track map)]—[13.Road Network Planning with Track Map]

3.1 Start Road Network Navigation

- Open a terminal on the host machine to start the vehicle's radar fusion node and odometer fusion

```
ros2 launch m3_bringup car_base.launch.py
```

- Open ros2kilted container terminal

```
exec_ros2kilted
```

- (RDK mainboard) Visualization is started on the accompanying virtual machine `Rosmaster-RoadNet-Ubuntu24.04`,

```
rviz2 -d /home/yahboom/code_ws/src/road_net_route/rviz/nav2_view.rviz
```

- Start road network planning navigation node in container terminal

```
ros2 launch road_net_route roadnet_nav2.launch.py
```

[!NOTE]

Parameter Description

map Grid map file name to be loaded, default value is `map.yaml`

pose_map Pose map file to be loaded. When navigation global positioning uses slam_toolbox pure positioning, the pose map must be passed in, default value is `map`

roadnet_file Road network description file, default value is `map.geojson`

3.2 Start Autonomous Driving

- Run in host machine terminal

```
ros2 launch m3_bringup auto_drive.launch.py combine_nav:=True unkeepplane:=True  
start_camera:=False
```

[!NOTE]

Parameter Description

- **combine_nav** Whether to combine road network navigation for autonomous driving, default is False. If True, the motion control of autonomous driving is provided by road network planning navigation, and traffic light response will directly replace the interruption and recovery acting on road network planning navigation

- `unkeeplane` Whether to close lane line detection model and thread, default is False. If road network navigation is started for autonomous driving, lane line detection and lane keeping functions will be automatically closed
- `start_camera` Whether to start camera node. If combined with AI large model, the camera node is started by the `multi_brains` launch file, not by this launch file

3.3 Start multi_brains Agent

Open a terminal on the vehicle and enter the command to start, or you can use the shortcut command `multi_brains`:

```
ros2 launch m3_bringup llm_agent_control.launch.py enable_route_nav:=True
```

```
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[model service-12] Cannot connect to server socket err = No such file or directory
[INFO] [model service-12]: process started with pid [116868]
[INFO] [action service usb-13]: process started with pid [116870]
[action service usb-13] [INFO] [1766401793.044243388] [action service usb]: enable_route_nav: True
[action service usb-13] [INFO] [1766401793.152994034] [action service usb]: ROS Action Service Initialization completed
[model service-12] [INFO] [1766401802.719665243] [model service]: Diffy LLM connection successful
[model service-12] [WARN] [1766401802.720780204] [rcl_logging_rosout]: Publisher already registered for provided node name. If this is due to multiple nodes with the same name then all logs for that logger name will go out over the existing publisher. As soon as any node with that name is destructed it will unregister the publisher, preventing any further logs for that name from being published on the rosout topic.
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear
[model service-12] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe
[model service-12] ALSA lib pcm_oss.c:397:(snd_pcm_oss_open) Cannot open device /dev/dsp
[model service-12] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model service-12] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream_open) Invalid card 'card'
[model service-12] ALSA lib confmisc.c:160:(snd_config_get_card) Invalid field card
[model service-12] ALSA lib pcm_usb_stream.c:482:(snd_pcm_usb_stream_open) Invalid card 'card'
[model service-12] ALSA lib pcm_dmix.c:1005:(snd_pcm_dmix_open) The dmix plugin supports only playback stream
[model service-12] Cannot connect to server socket err = No such file or directory
[model service-12] Cannot connect to server request channel
[model service-12] jack server is not running or cannot be started
[model service-12] JackShmReadWritePtr::~JackShmReadWritePtr - Init not done for -1, skipping unlock
[model service-12] JackShmReadWritePtr::~JackShmReadWritePtr - Init not done for -1, skipping unlock
[model service-12] [INFO] [1766401803.247590096] [ASR_Detect]: paraformer-realtime-v2 ASR Engine initialized successfully
[model service-12] [INFO] [1766401803.262672610] [model service]: Tongyi TTS Engine initialized successfully
[model service-12] [INFO] [1766401803.263389574] [model service]: ROS LLM Service Initialization completed
```

3.4 Demonstration Case

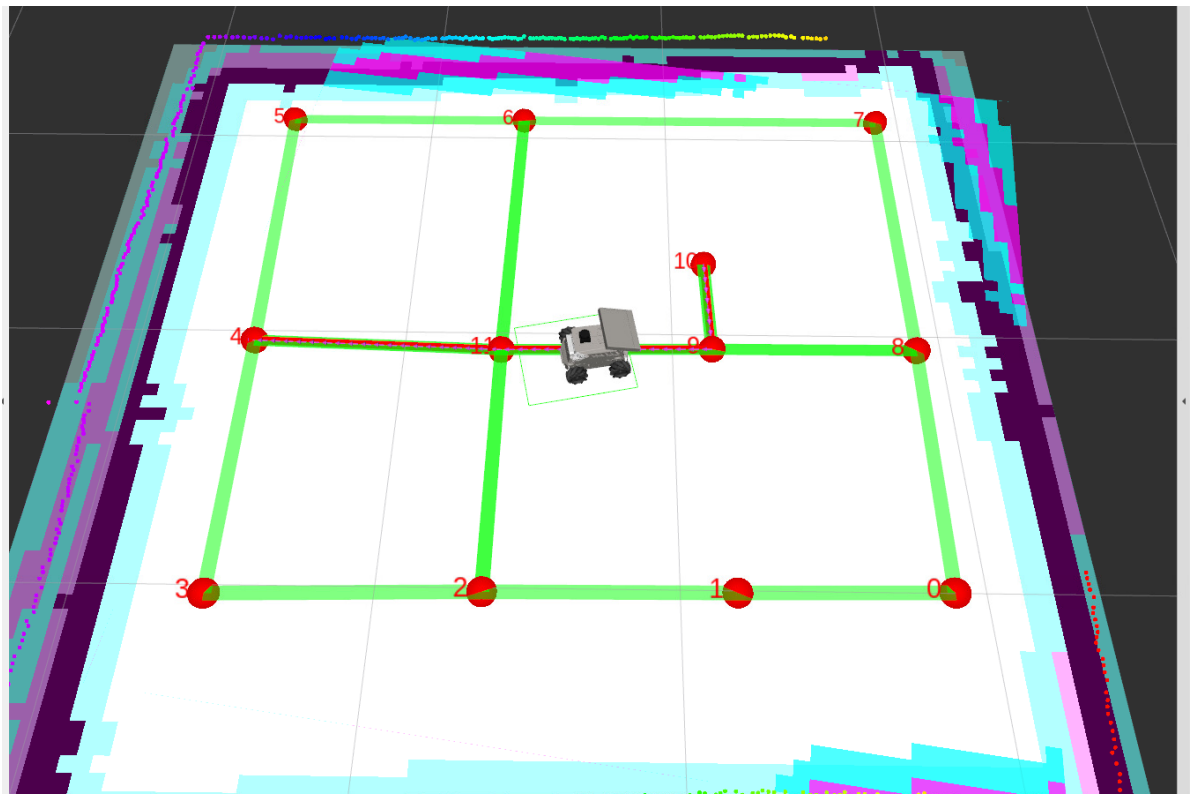
- Navigate to the construction site while obeying traffic rules

After waking up the vehicle with "Hello, yahboom", when the vehicle responds and the recording prompt sound "dang—" appears, give the command to the vehicle

```
jetson@yahboom: ~/yahboomcar_ros2_ws/yahboomcar_ws
jetson@yahboom: ~/yahboomcar_ros2_ws/yahboomcar_ws 92x30
[asr-16] [INFO] [1764657605.668674285] [asr]: 1
[asr-16] [INFO] [1764657605.729629067] [asr]: 1
[asr-16] [INFO] [1764657605.790110264] [asr]: 1
[asr-16] [INFO] [1764657605.849843115] [asr]: -
[asr-16] [INFO] [1764657605.911233176] [asr]: -
[asr-16] [INFO] [1764657605.972604740] [asr]: -
[asr-16] [INFO] [1764657606.033191989] [asr]: -
[asr-16] [INFO] [1764657606.093282964] [asr]: -
[asr-16] [INFO] [1764657606.153825827] [asr]: -
[asr-16] [INFO] [1764657606.213770717] [asr]: -
[asr-16] [INFO] [1764657606.274789948] [asr]: -
[asr-16] [INFO] [1764657606.335223463] [asr]: -
[asr-16] [INFO] [1764657606.395954556] [asr]: -
[asr-16] [INFO] [1764657606.458824763] [asr]: -
[asr-16] [INFO] [1764657606.518130303] [asr]: -
[asr-16] [INFO] [1764657606.577540327] [asr]: -
[asr-16] [INFO] [1764657606.640408294] [asr]: -
[asr-16] [INFO] [1764657606.698531713] [asr]: -
[asr-16] [INFO] [1764657606.757904103] [asr]: -
[asr-16] [INFO] [1764657606.818232686] [asr]: -
[asr-16] [INFO] [1764657606.878078981] [asr]: -
[asr-16] [INFO] [1764657606.939070019] [asr]: -
[asr-16] [INFO] [1764657606.998873624] [asr]: -
[asr-16] [INFO] [1764657607.060230339] [asr]: -
[asr-16] [INFO] [1764657607.120183356] [asr]: -
[asr-16] [INFO] [1764657608.765174546] [asr]: 导航去工地途中遵守交通规则。
[asr-16] [INFO] [1764657608.765862122] [asr]: 😊okay, let me think for a moment...
[model_service-14] [INFO] [1764657610.791974513] [model_service]: 决策层AI规划:1. 导航去工地
[model_service-14] 2. 途中遵守交通规则

[action_service_nuwa-15] [INFO] [1764658284.847215058] [action_service]: Published message:
机器人反馈:执行control_sign(enable)完成
[action_service_nuwa-15] [INFO] [1764658284.858613098] [action_service]: Published message:
机器人反馈:执行['navigation(D)', 'control_sign(enable)']完成
[model_service-14] [INFO] [1764658286.053808441] [model_service]: "action": [], "response":
交通规则已开启, 现在安全出发去工地
[model_service-14] [INFO] [1764658287.629069342] [model_service]: "action": ['finishtask()'],
"response": 已到达工地, 任务完成!
[action_service_nuwa-15] [INFO] [1764658291.935198082] [action_service]: Published message:
机器人反馈:回复用户完成
[model_service-14] [INFO] [1764658293.260168282] [model_service]: "action": ['finishtask()'],
"response": 已到达工地, 任务完成!
```

- The vehicle will then follow the road network planned route to the target point "Construction Site"



- In AI large model combined with road network autonomous driving, you can also let AI not respond to traffic signals. You can add conditions to the command: ignore traffic signals, at which time AI will temporarily close the response to traffic signs.